An updated systematic review and dose-response meta-analysis of protein intakes effects upon lean soft tissue mass, muscle, strength, and function in healthy adults*

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Abstract

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Introduction

In a recent systematic review and meta-analysis Nunes et al. (2022) examined the dose-response effects of protein intake relative to body mass upon lean soft tissue mass, muscle mass/size, strength, and function both with and without concomittant resistance training (RT). However, whilst they examined both categorical protein intakes ('<1.2 g/kg/day', '1.2–1.59 g/kg/day' and ' 1.6 g/kg/day') as well as continuous (as g/kg/day) in the the latter only linear functional forms were utilised which imply a constant slope of proteins effect across varied intakes. It may be the case that some kind of non-linear functional form may better describe the dose-response relationship. For example, in earlier work Morton et al. (2018) used a piecewise linear model and estimated a so-called "breakpoint" where a plateau in protein intakes effects might occur for lean soft tissue (breakpoint = 1.62 g/lg/day [95%CI: 1.03, 2.20]) . Tagawa et al. (2021; 2022) have also reported possible non-linear functional forms where they utilised piecewise splines though inflexion points were selected arbitrarily; 1.3 g/kg/day for strength outcomes without RT and for lean soft tissue mass with RT, and 1.5 g/kg/day for strength outcomes with RT). However, both approaches are somewhat limited in specifically forcing a single "breakpoint". Instead, more flexible modelling strategies using extensions of generalised additive models to meta-analytic scenarios may allow for clearer determination of the dose-response function based on existing studies data. Further, the use of singular models renders dose-response estimates conditional on that particular model (e.g., a linear model can only produce a linear dose response function, or the inclusion of a singular breakpoint will only allow the model to "flex" about that point).

Whilst the models previously explored may have plausibility, at least over the range of data available, these previous analyses have also not made any attempts at comparing functional forms implied by different

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models. However, even comparing amongst different candidate models with regards to the relative evidence for and against different functional forms as being the underlying data generating process has limitations. Any predictions are conditional on the candidate model selected and selection process. Indeed, when fitting models to data, even when all are plausible candidates, there is uncertainty in the possible models that could be fit and the selection process. However, with Bayesian approaches to modelling it is simple to combine, in a weighted fashion determined by their relative evidence, estimates from all models examined and directly incorporate this uncertainty (Hinne et al., 2020). Adopting a Bayesian approach for evidence synthesis also allows the incorporation of prior information that might be relevant to inference. For example, we also have precise estimates of the effects of RT without additional protein intervention from previous meta-analyses for lean soft tissue mass, muscle mass/size, strength, and functional outcomes (Kirk et al., 2024; Steele et al., 2023). Most, though not all, previous protein intake intervention studies have examined RT as a concomitant intervention. As such we can also utilise these estimates, in addition to those from non-RT control groups, to set informative priors for the expected effect sizes at habitual protein intakes (i.e., the typical protein intakes of people taking part in intervention research examining RT without a concomitant intervention to increase protein intake).

Previous meta-analyses examining the effects of protein intake upon the outcomes noted have typically not pre-registered a smallest effect size of interest range to draw conclusions regarding the meaningfulness of the effects identified. Here we also set the smallest effect size of interest to interpret the how meaningful the effects of increased protein intake are upon outcomes. This would enable both current recommendations to be updated if necessary and also generate a functional form enabling more precise theoretical predictions of effects that future studies might offer severe tests of i.e., predicted magnitudes of condition by time interaction effects for comparing different doses of protein in a randomised trial.

Thus, we sought to revisit the studies that were included in the review of Nunes et al. (2022), in addition to updating searches since the date of their last search to include any newer studies replicating and extending the systematic searches conducted previously. Our aim was to explore the dose-response relationships between experimental manipulation of protein intake and both lean soft tissue mass, muscle, strength, and function in healthy adults.

Methods

This systematic review was originally pre-registered in July 2025 on the Open Science Framework following the structure for registration with the International Prospective Register of Systematic Reviews (PROSPERO). The pre-registration and any updates to it is available at: https://osf.io/w2qpx/overview. We followed the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) guidelines (Page et al., 2021).

Updates to pre-registration

Some minor ammendments were made to our analysis plan after pre-registeration when we realised that protein intake as a predictor for the arm level within studies is also estimated with error (i.e., sampling error) where it is a combination of dietary intake (experimentally manipulated or not) and any experimentally manipulated supplementation to increase protein intake. As such, we updated our approach and extracted sufficient data from all included studies to enable us to either use the reported standard errors for protein intake in g/kg/day, or to propagate error in the calculation of protein intake, and to then model this directly using latent error models for the predictor. In addition a typo in the priors was noted where the scale for weakly regularising priors on coefficients other than the effects of RT was set to 10 when it should have been 1 given the former allows for completely implausible standardised mean changes Lastly, instead of the slopes at different protein intakes for the magnitude of the median difference between intervention and control arms, we have updated to consider the paired contrasts across the dose-response function for different protein intakes in contrast with the recommended daily allowance of 0.8 g/kg/day as the reference condition.

Search Strategy

The search strategy used by Nunes et al. (2022) as replicated exactly with searches and screening being conducted by two investigators (EAN, ML) with any disagreements planned to be resolved by a third investigator (JS) though no disagreements occured. The original search strategy can be seen in the online supplementary materials for Nunes et al. (2022) here. Searches had previously been conducted up to September 2020 by Nunes et al. (2022) and so we performed searches from this date up to July 2025. In addition, any papers not identified in the original or updated searches but which were known to the author team and met inclusion criteria were also identified and included. Rayyan (Ouzzani et al., 2016) was used to manage the screening process after searches were conducted.

Inclusion/exclusion criteria

We will examined studies including healthy adults and which manipulated protein intake experimentally with or without inclusion of a concomitant RT intervention performed in any settings. The following population, intervention, comparator, and outcomes (PICO) was used to guide inclusion:

- Population
  - Adult participants (healthy) aged 18 years or older.
- Intervention(s) or exposure(s)
  - Additional protein ingestion with or without the addition of resistance exercise (increasing protein ingestion as the primary intervention using nutritional supplements or diet at any dose or level).
- Comparator(s) or control(s)
  - Placebo or no intervention (control).
- Outcomes
  - Lean soft tissue mass (or a closely related measure i.e., fat-free mass; Heymsfield et al. (2024)), direct measures of muscle mass/size (both whole muscle and muscle fiber), muscle strength and performance in tests of physical function.

Studies including participants with decreased mobility, frailty, obesity, or any chronic or infectious diseases were not included. Also, interventions were carefully screened for the presence of any potential active ingredient that might impact lean soft tissue mass gain other than protein (i.e. creatine, phosphatidic acid, omega-3 fatty acids, anabolic steroids, and beta-hydroxy-beta-methylbutyrate [HMB]) or if they were intended to cause weight loss (e.g., energy restriction). If a particular study testing potential active supplements had intervention groups fitting our inclusion criteria, the study was included and assessed to extract the respective arm based data of protein and control/placebo interventions only. No restrictions were imposed regarding additional protein dose or the duration of the intervention protocol. Only randomised controlled trials using parallel arm designs (i.e., where arms where independent) were included.

For primary outcomes we considered the following as valid operationalisations of these constructs: for whole body lean soft tissue mass, dual-energy X-ray absorptiometry (DEXA), hydrodensitometry, bioimpedance analysis (BIA), skinfolds, and/or whole-body air plethysmography (BodPod®); for direct measures for muscle size/mass, magnetic resonance imaging (MRI), computed tomography (CT), ultrasound (US), appendicular measures of muscle mass using DEXA or BIA, and any muscle biopsy measures for fiber cross sectional area; for strength, any repetition-maximum (isotonic), maximal isometric, or maximal isokinetic testing for all tasks/exercises; and for physical function, timed up and go (TUG), chair-based testing (i.e., sit to stand), sit-to-stand tests, gait speed tests, balance tests, short physical performance battery tests, stair climb tests, time or distance-limited walking tests, jump tests, agility tests, flexibility tests, and any other tests involving activities of daily life.

Data extraction/coding

It was originally planned that data would be extracted by at least two investigators after random allocation of newly identified studies with discrepancies checked by a third (JS). However, two members of the original extraction team were unable to complete their extraction and so two authors (LR, AVM) extracted the studies allocated to them with a third (JS) going back through all included studies, including re-extracting

data from the original studies included in Nunes et al. (2022) in addition to all newly identified studies, such that all studies had been checked and extracted by at least two authors.

For each study we extractedd bibliometric data including pubmed ID, DOI, authors, corresponding author email, journal, year of publication, title of article, and how many arms from the study were to be included in our analysis. For each arm within each study data was extracted participant characteristics, including location (i.e., means) and scale (i.e., standard error, or standard deviations) for any continuous variables, which included sex (proportion male), age and body mass, sample size at baseline, intervention characteristics including protein source and what the placebo arm consisted of, whether concomitant RT was performed, and dietary protein intake (at baseline and during the study i.e., after intervention). On an arm by arm basis, and for each time point reported in the study, all whole body lean soft tissue mass and direct muscle mass/size outcomes, strength outcomes, and function outcomes in addition to the outcome group they belong to (whole body lean soft tissue mass; direct muscle size/mass; strength; function), the measurement method, the units of measurement, samples sizes, and means and either standard deviations, standard errors, or confidence intervals were extracted in addition to contrasts between each time points mean deltas (e.g., for two timepoints 1 and 2, timepoint 2 – timepoint 1) and their standard deviations or standard errors, t statistics, p values, confidence intervals, and correlation between timepoints. If not available in tables or the text, data was extracted from figures using the online tool WebPlotDigitizer (all data extracted from figures is available in the online supplementary materials). Authors were contacted for missing data. We originally planned to contact all authors to request individual participant level data, however utlimately opted not to pursue this due to resource and time constraints for the author team to commit to this.

Risk of bias and quality of evidence assessment

The RoB2 tool (Sterne et al., 2019) was used to examine risk of bias in the included studies, both those included in Nunes et al. (2022) and all newly identified studies. Further the quality of evidence was appraised using the GRADE criteria.

Statistical analysis

Statistical analysis of the data extracted was be performed in R (v 4.4.3; R Core Team, https://www.r-project.org/) and RStudio (v 2026.07.0; Posit, https://posit.co/). All of the code utilized for the data preparation and analyses is available on the Open Science Framework page (https://osf.io/37bzv/ and the corresponding GitHub repository (https://github.com/jamessteeleii/protein_dose_response_meta).

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We cite all software and packages used in the analysis pipeline using the `grateful` package [@rodrigue
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Standardised mean changes with small sample bias correction were calculated within each arm and for each outcome for each contrast relative to baseline using Becker's (1988) d (i.e., where the numerator is the mean for a given follow-up timepoint, n, after baseline minus the mean for baseline; timepoint 0+n - timepoint 0; where only mean delta contrasts between timepoints are reported, though baseline standard deviations are available, we used the delta score for the contrast to baseline as the numerator). However, given we are including only randomised controlled trials we will use the pooled baseline standard deviations as the denominator, akin to Morris' (2008) PPC_2, given that participants come from the same population at randomisation. We used standardised mean change scores due to the varying outcome operationalisations for the constructs considered (i.e., in both specific method of observation and in the units of measurement) that were extracted. Further, given the nature of the constructs of interest this facilitated comparison with other large scale meta-analytic estimates (e.g., Steele et al. (2023)) to enable interpretation of the magnitude of effects identified.

Though we have specific criteria for the selected models described below that were fit, our analysis should be considered exploratory and aimed at principled dose-response model selection. Our dependent variable in all models is the standardised mean change relative to baseline at each follow-up timepoint calculated as described above. Analysis was aimed at effect estimation within a Bayesian meta-analytic framework (Kruschke & Liddell, 2018). For all analyses, the model estimates and their precision, along with conclusions based upon them, are interpreted continuously and probabilistically, considering the data quality, the

plausibility of effect, and the previous literature, all within the context of our models.

We originally planned to use the main package `brms` (Burkner, 2018) in fitting all the Bayesian meta-analysis models described below. We still report the results of these models as our primary analysis. However, as noted above we subsequently realised that our primary predictor of protein intake is one estimated with error (i.e., sampling error) for each study and arm included. As such, we updated our approach and extracted sufficient data from all included studies to enable us to either use the reported standard errors, or to propagate error in the calculation of protein intake, and to then model this directly using measurement error models for the predictor. Therefore, all models remain the same, or as closely as possible, in terms of their described function forms if not exactly in their parameterisation, for fitting and inclusion of covariates and random effects, but instead the `brms` models were translated to Stan models in order to fit them using `cmdstanr` with this latent error in variables approach. Essentially this treated each estimate of protein intake as an estimate with error and for which the underlying true protein intake was modelled as a latent parameter and it was this which was treated as the predictor within each higher model. These errors in variables models are reported in the supplementary materials. We used four Monte Carlo Markov Chains each with 2000 warmup and 6000 sampling iterations. Note this deviated from our pre-registration in order to reduce the time required for model fitting given that we were primarily interested in estimation rather hypothesis testing and so did not calculate Bayes Factors. Trace plots were produced along with rhat values to examine whether chains were converged for each model, and posterior predictive checks for each model were also examined to understand the model implied distributions. These were examined across all models prior to comparisons to ensure good convergence, that all rhat values were close to 1, and posterior predictive checks seem appropriate distributions for the observed data. Prior and posterior draws for both global grand means of marginal predictions and marginal contrasts for protein intakes at intervals of 0.2 g/kg/day with 0.8 g/kg/day set as the reference intake and contrasts with this up to 4.4 g/kg/day (i.e., the highest protein intake point estimate reported in the included studies).

Data was hierarchical across three levels (i.e., effects within arms within studies), and so, we included random (i.e., cluster) intercepts using implicit nested coding across these levels in all models. We then examined a selection of models for the functional form of dose-response effect of protein intake. For each model, protein intake was included as a fixed (i.e., population) effect in g/kg/day and centred at the median value for protein intake across all control and placebo arms i.e., considered to be the habitual protein intake, which was 1.1 g/kg/day. In all models we included a dummy coded fixed effect for whether the study included a RT intervention (code = 0) or not (code = 1) and set informative priors based on the effect estimates for strength, lean soft tissue mass and muscle mass/size, and functional outcomes based data from previous meta-analyses using similar arm based models as those described (Kirk et al., 2024; Steele et al., 2023). These priors reflect the typical effect magnitudes for these outcomes under either RT or non-RT conditions with habitual protein intakes. Priors for all other fixed effect coefficients were set to be weakly regularising using a student t distribution with 3 degrees of freedom, mu of 0, and sigma of 1 permitting a reasonable range of prior values of standardised mean effects. Priors for random intercepts followed a half student-t prior with 3 degrees of freedom and a scale parameter set to 2.5 centred at a mu of 0.

We began by fitting a selection of possible dose-response functional forms for the effects of protein intake in g/kg/day as follows:

1. Intercept only model i.e., a model assuming no dose-response effect of protein intake
2. Linear model i.e., a model assuming a linear dose response effect of protein intake
3. Linear splines with knot at 1.6 g/kg/day i.e., a model assuming linear dose-response effects that differ in magnitude below or above a protein intake of 1.6g/kg/day (similarly to previous "breakpoint" models)
4. Nonlinear smooth model using splines[a] i.e., a model assuming a dose-response that is a non-linear smooth function of protein intake with three basis functions

Each of these describes the functional form for the dose-response of protein intake upon standardised mean change from baseline in outcomes and vary in the extent to which they restrict the shape of the function[b].

[a]Note, for the `brms` models we originally pre-registered fitting thin plate regression splines and continue with this approach here for our primary analysis. However, for the Stan models due to problems with convergence and parametrisation, we opted to fit natural cubic splines albeit similarly with three basis functions.

[b]Note, we originally planned to also examine a model assuming dose-response effects are a function of both a linear and

We then fit each of these functional forms across four groupings of models varying in parameters included:

1. Models with only fixed (i.e., population) level effects for protein intake
2. Models with only fixed (i.e., population) level effects for protein intake, and also including fixed effects for intervention duration in weeks (centred on 12 weeks), and age in years (centred on 25 years)
3. Models with both fixed (i.e., population) level effects for protein intake in addition to random (i.e., cluster) level effects for protein intake at the study level
4. Models with both fixed (i.e., population) level effects for protein intake in addition to random (i.e., cluster) level effects (slopes or smooths) for protein intake at the study level, and also including fixed effects for intervention duration in weeks (centred on 12 weeks), and age in years (centred on 25 years)

The exception being the intercept only model which of course cannot be fit with random slopes or smooths for protein intake.

Lastly, we also fit all the above models with the inclusion of a fixed interaction term (either linear, or fixed factor smooths by each level) between protein intake and whether the study included a RT intervention to include the possibility that protein intake requirements may differ if participating in resistance training.

As such, a total of 26[c] models varying in complexity regarding the functional forms for dose-response effects of protein intake, the inclusion of additional study and arm level moderators, whether or not the dose-response effects of protein intake are modelled as varying between studies, and whether or not the effects of protein intake varying with and without the concomitant inclusion of a RT intervention were fit and compared in terms of the evidence supporting them relative to one another as the data generating process. The included moderators of intervention duration and age were chosen to be centred on the typical duration (12 weeks) and age for participants (25 years) in studies such that the model predictions and contrasts are interpretable as being at these values.

Criteria for conclusions / inference criteria

As noted, we utilise Bayesian model averaging across the set of candidate models noted above in order to account for uncertainty in the model selection process. Across these models we used leave-one-out cross-validation (LOO-CV) to estimate the expected log predictive density (ELPD) and calculate weights using the stacking approach (Vehtari et al., 2017; Yao et al., 2018). We then took posterior draws of the global grand means (i.e., the expectation of effect size, without the inclusion of random effects) for predictions and contrasts across different values of protein intake and calculate the means with 95% quantile intervals. These draws for each model were then be combined using the LOO-CV weights to generate our Bayesian model averaged predictions and contrasts.

We collectively agreed as an author group upon the smallest effect size of interest on the standardized mean change (−0.1, 0.1) scale and thus set this as our region of practical equivalence (ROPE). This allows us to also examine the probability that the contrasts for protein intakes above the reference value of 0.8 g/kg/day might produce a meaningful positive effect (i.e., greater than the smallest effect size of interest) in comparison to this reference by examining the mass of the posterior distribution exceeding the upper limits of the ROPE, and also the percentage of the posterior distributions mass that was within the ROPE, thus reflecting the probability of practically equivalent effects. For reference and to aid interpretation for the reader, this ROPE equates to a lean mass change of ~0.534 kg, a fat free mass change of ~0.604 kg, a muscle thickness change (for a muscle of pre-test 3.62 cm thickness) of ~0.0548 cm, and a squat, bench press, and deadlift one repetition maximum change of ~5.48 kg, ~4.32 kg, and ~5.7 kg respectively. These values for reference are based on meta-analysis of the baseline lean mass and fat free mass standard deviations from another recent systematic review (Benito et al., 2020), meta-analysis of baseline anterior thigh muscle thickness standard deviations adjusted for mean values (due to different sites of measurement) based on presented in a recent review (Buckner et al., 2023), and the standard deviations for bench press and squat

logarithmic transformation of protein intake to allow a parametrised non-linear function permitting a plateau, but the use of priors that were based on habitual protein intake required centring of the protein intake meaning we could not log transform the predictor given zero or negative signed values. However, we felt the inclusion of the thin plate spline regression model enabled sufficient flexibility to allow for this kind of functional form to emerge.

[c]The pre-registration contains a miscounting error and states 28 models.

exercises (filtered to the first competition, unequipped, and drug tested) from the open powerlifting dataset (see https://openpowerlifting.gitlab.io/opl-csv/bulk-csv.html).

Results

Discussion

Conclusion

Intakes up to ~3 g/kg/day do not meaningfully improve outcomes compared to 0.8 g/kg/day. Above this uncertainty in estimates renders difficulty in determining whether effects are meaningful or not with x-y% probability of positive meaningful effects and x-y% probability of practically equivalent effects.

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